LARGE LANGUAGE MODELS: PRODUCTION AND DEPLOYMENT

# Fine-Tuning Process and Types of Fine-Tuning

From a Pre-trained LLM to a Specialized Application Model

Lecture 3 | 60 minutes | Process → Taxonomy

TOPICS COVERED IN THIS LECTURE

- Experimental Pipeline
- Measurable Task & Baseline
- Base Model Selection
- Dataset Quality
- Train / Validation / Test
- Data Leakage
- The Training Loop
- Loss & Backpropagation
- Fine-Tuning Taxonomy
- Full vs Selective vs PEFT



PART A

# The Fine-Tuning Process

From the decision to fine-tune to a deployed specialized model — **fine-tuning is not simply calling train().**

IN THIS SECTION

01 Where Are We?
02 Running Example
03 Experimental Pipeline
04 Task & Baseline
05 Base Model
06 Dataset Building
07 Splits & Leakage

SECTION 01 · ORIENTATION

Quick Recall

# Where Are We? — and Quick Recall

## WHERE ARE WE?

Pre-trained LLM

↓

Decide that Fine-Tuning is Needed

↓

**TODAY**

**How do we fine-tune correctly?**

↓

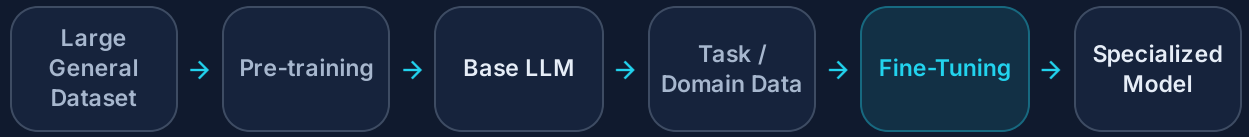
What different forms can adaptation take?

**Fine-tuning is not simply calling train().**

## QUICK RECALL: WHAT IS FINE-TUNING?

> Fine-tuning is the **continuation of training** of an already pre-trained model using a smaller task-specific or domain-relevant dataset.

### THE PIPELINE AT A GLANCE




**PRE-TRAINING**

→ broad / general capability

**FINE-TUNING**

→ specialization

SECTION 01 · RUNNING EXAMPLE

Running Example

# Running Example: University Helpdesk Assistant

One example carried through the entire lecture.

USER QUERY

“My semester fee was deducted from my bank account, but the portal still shows unpaid.”

REQUIRED SYSTEM RESPONSE

```
Department: Accounts
Category: Payment Reconciliation
Priority: High
Action: Verify transaction reference
```

**Q1** Does the original LLM understand the English sentence?

↳ A1 Probably yes.

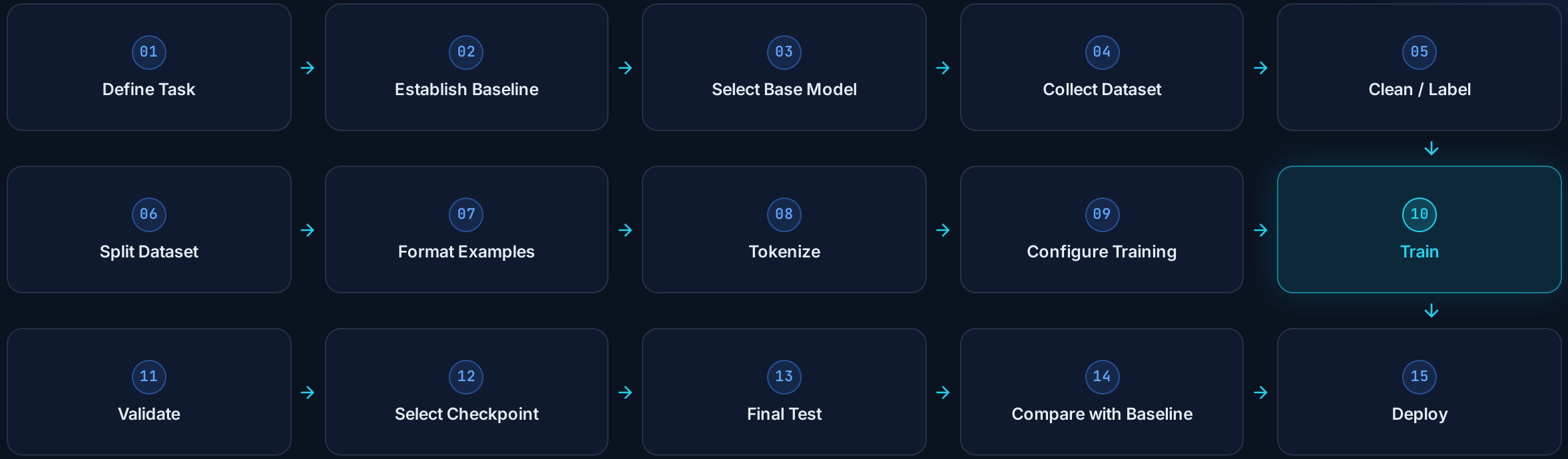
**Q2** So what are we teaching?

↳ A2 **The application’s required task-specific behavior.**

SECTION 01 · THE PIPELINE

The Pipeline

# Fine-Tuning Is an Experimental Pipeline




Training is only one stage of fine-tuning.

SECTION 01 · STEPS 1–2

Task · Baseline

# Steps 1–2 · Define a Measurable Task & Establish a Baseline

## STEP 1 · DEFINE A MEASURABLE TASK

**POOR**

"Make my LLM better."

**BETTER**

"Classify student queries into 10 predefined departments."

**STRONGER**

"Classify student queries into 10 departments while maintaining valid structured output and achieving the required macro-F1."

If better is not measurable, you cannot prove that fine-tuning worked.

## STEP 2 · ESTABLISH A BASELINE

Before fine-tuning, measure how far the base model already gets.

```
Base Model + Best Reasonable Prompt → Evaluation
```

| | |
|---|---|
| CLASSIFICATION | Accuracy · Precision · Recall · F1 · Confusion matrix |
| STRUCTURED GEN. | schema validity · required-field accuracy · exact match where appropriate |
| PRODUCTION | latency · throughput · memory · cost |

```
Base model macro-F1 = 0.67 → Fine-tuned macro-F1 = 0.91
```

No baseline → no evidence of improvement.

SECTION 01 · STEP 3

Base Model

# Step 3 · Choose the Base Model

## CONSIDERATION CHECKLIST

- ✓ model quality
- ✓ parameter count
- ✓ context length
- ✓ supported languages
- ✓ domain capability
- ✓ license
- ✓ hardware requirement
- ✓ fine-tuning support
- ✓ deployment target
- ✓ inference cost

## A SELECTION DECISION

`3B model` → **93%** task accuracy

`70B model` → **94%** task accuracy

**Should we automatically deploy the 70B model?** **No.**

One point of accuracy is rarely worth multiples of memory, latency, cost and operational complexity.

**Largest model ≠ best production model.**

SECTION 01 · STEP 4

Dataset

# Step 4 · Build the Dataset

## EXAMPLE DATA

| USER INPUT | EXPECTED RESULT |
| --- | --- |
| “My fee payment failed.” | ACCOUNTS |
| “My hall ticket is unavailable.” | EXAMINATION |
| “I lost my ID card.” | STUDENT_SERVICES |
| “How do I get a transcript?” | EXAMINATION |

**A useful dataset needs:**

correct examples · representative examples · sufficient diversity · realistic phrasing · consistent labels · adequate coverage

## DATA QUALITY DECIDES THE SIGNAL

| Good supervision | Incorrect supervision |
| --- | --- |
| ↓ | ↓ |
| Useful learning signal | Wrong learning signal |
| ↓ | ↓ |
| Better generalization | Poor behavior |

Input: “My ATM card was stolen.”

Incorrect target: PAYMENT_FAILURE

**What will the model learn?**

It will be explicitly encouraged to learn the wrong mapping.

SECTION 01 · DATASET QUALITY

Dataset

# Consistency & Diversity

## CONSISTENCY

BAD LABELS — ONE INTENT, THREE TAGS

- “My card is stolen.” → LOST_CARD
- “My debit card was stolen.” → FRAUD
- “I lost my ATM card.” → CARD_MISSING

*...unless those distinctions are intentionally defined.*

All equivalent cases → LOST_CARD

**Your annotation policy must be clearer than the behavior you expect from the model.**

## DIVERSITY

- “My payment failed.” →
- “Payment isn’t going through.” →
- “Why can’t I pay?” →
- “The transaction keeps failing.” →
- “Unable to complete payment.” →

→ PAYMENT_FAILURE

‘We want the model to learn the concept, not memorize one sentence.’

# Train / Validation / Test — and Data Leakage

DATASET

- 80% **TRAIN** — learn model parameters
- 10% **VALIDATION** — select model / configuration
- 10% **TEST** — final unbiased evaluation

80 / 10 / 10 is only an example — not a universal rule.

## DATA LEAKAGE

TRAIN
“My ATM card has been stolen.”

=

TEST
“My ATM card has been stolen.”

If the model scores correctly, are we measuring generalization? → **Not necessarily.**

*“A student receives the final examination paper before the examination and gets 100%.”*

**DEFINITION –** Data leakage occurs when information that should remain outside training improperly influences training or model selection.

SECTION 02

# Inside the Training Loop

What actually happens when we press train — from formatted examples to parameter updates.

IN THIS SECTION

- Fine-Tuning Record
- Chat Format
- Tokenization
- Forward Pass
- Loss
- Backpropagation
- Optimizer & Learning Rate
- Batches & Epochs
- Overfitting
- Checkpoints

SECTION 02 · FORMATTING

FORMAT & TOKENIZE

# Formatting a Fine-Tuning Example

## BASIC SUPERVISED RECORD

Instruction: Classify the customer complaint.

Input: My debit card has been stolen.

Expected Output: **LOST_CARD**

Instruction + Input + Desired Output

## CHAT / INSTRUCTION FORMAT

**SYSTEM** "You are a university helpdesk assistant."

defines general behavior

**USER** "My tuition payment is not reflected."

provides the input/request

**ASSISTANT** Department: Accounts / Priority: High / Action: Verify transaction reference.

the response the model should learn to generate

Training Example → Tokenizer → Token IDs → Attention Mask → Labels → **Model**

The model does not train directly on English strings. **It operates on numerical token representations.**

# The Training Loop

NEXT BATCH

01
**Training Example**
formatted, tokenized input

02
**Forward Pass**
pure computation

03
**Prediction**
output probabilities

04
**Compare with Desired Output**
prediction vs target

05
**Loss**
a number to minimize

06
**Backpropagation**
propagates the loss backwards

07
**Gradients**
one per trainable parameter

08
**Optimizer**
applies the update

09
**Parameter Update**
`θ_new = θ_old - η ∇L`

repeats for every batch

repeats for every epoch

This loop is the whole of training — every remaining slide in this section zooms into one arrow of this diagram.

SECTION 02 · TRAINING LOOP

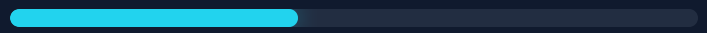
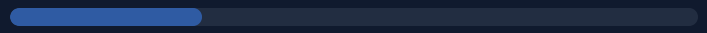
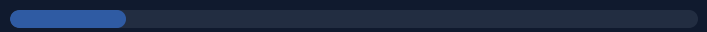
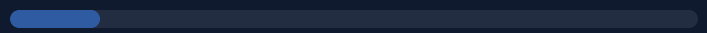
FORWARD · LOSS

# Forward Pass & Loss

## FORWARD PASS

Input: “My hall ticket is unavailable.”

PREDICTION PROBABILITIES — CLASS · P

| | | |
|---|---|---|
| EXAMINATION |  | 0.42 |
| ACCOUNTS |  | 0.28 |
| LIBRARY |  | 0.17 |
| ADMISSION |  | 0.13 |

EXAMINATION wins — at only 0.42

Tokens → Model → Prediction

STAGES 02–04 · PURE COMPUTATION · WEIGHTS UNTOUCHED

No parameter update has occurred yet.

## Σ LOSS

TARGET: EXAMINATION

vs

PREDICTION: EXAMINATION = 0.42

Loss quantifies **disagreement** between prediction and target.

↗ Poor prediction → **larger loss**

↘ Better prediction → **smaller loss**

” Loss gives the training process a **numerical objective to improve**.

SECTION 02 · TRAINING LOOP

BACKPROP · OPTIMIZER

# Backpropagation, Optimizer & Learning Rate

## BACKPROPAGATION

LOOP STAGES 05–07

05 **Loss**
the number to minimize

↓

06 **Backpropagation**
propagates the loss backwards

↓

07 **Gradient**
for each trainable parameter

> A gradient tells us approximately: **how would the loss change if this trainable parameter changed?**

We do not derive the calculus in this lecture.

## OPTIMIZER UPDATE

LOOP STAGES 08–09

$$\theta_{new} = \theta_{old} - \eta \nabla L$$

$\theta$ = parameter · $\nabla L$ = gradient · $\eta$ = learning rate

Very small learning rate → **slow progress**

Very large learning rate → **unstable updates**

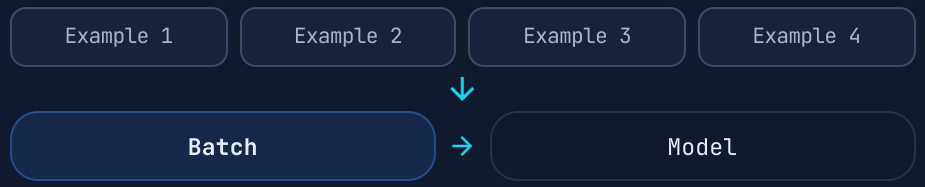
Suitable learning rate → **useful progress**

DETAILED TUNING COMES IN LECTURES 7–8

# Batches, Epochs & Overfitting

## BATCH

A subset of training examples processed together.




## EPOCH

One complete pass through the training dataset.

1,000-EXAMPLE DATASET

- **Epoch 1** → model sees all 1,000
- **Epoch 2** → sees them again
- **Epoch 3** → again

Are more epochs always better? → **No.**

## OVERFITTING

| | | |
|---|---|---|
| TRAINING | ↑ ↑ ↑ | performance keeps rising |
| VALIDATION | ↗ ↑ ↓ | performance rises, then falls |

↘ WHERE THEY DIVERGE = OVERFITTING

The model becomes very good on training examples but **fails to generalize** to unseen examples.

SIGNS

- ⚠ train loss continues decreasing
- ⚠ validation loss increases
- ⚠ training accuracy very high
- ⚠ test performance weaker

> "Memorizing practice questions instead of learning the concept."

SECTION 02 · TRAINING LOOP

CHECKPOINTING

# Checkpoints, Validation & Final Test

## DURING TRAINING

PHASE 01

Epoch / Step → Checkpoint → Validation

### CHECKPOINT SCORES ON VALIDATION

| | | |
|---|---|---|
| Checkpoint 1 | | F1 = 0.84 |
| **Checkpoint 2** | BEST | F1 = 0.91 |
| Checkpoint 3 | | F1 = 0.89 |

! The final training checkpoint is **not automatically the best model**.

THEN

## AFTER MODEL SELECTION

PHASE 02

**Best Validation Checkpoint**
selected by validation score

↓

**Previously Unseen Test Set**
touched exactly once

↓

**Final Performance**
the number you report

⚠ The test set should **not repeatedly drive tuning decisions**.
REPEATED PEEKING QUIETLY TURNS IT INTO A TUNING SET

SECTION 02 · EVALUATION

RESULTS

# Base vs Fine-Tuned Model

THIS COMPARISON IS ONLY POSSIBLE BECAUSE STEP 2 DEFINED THE BASELINE

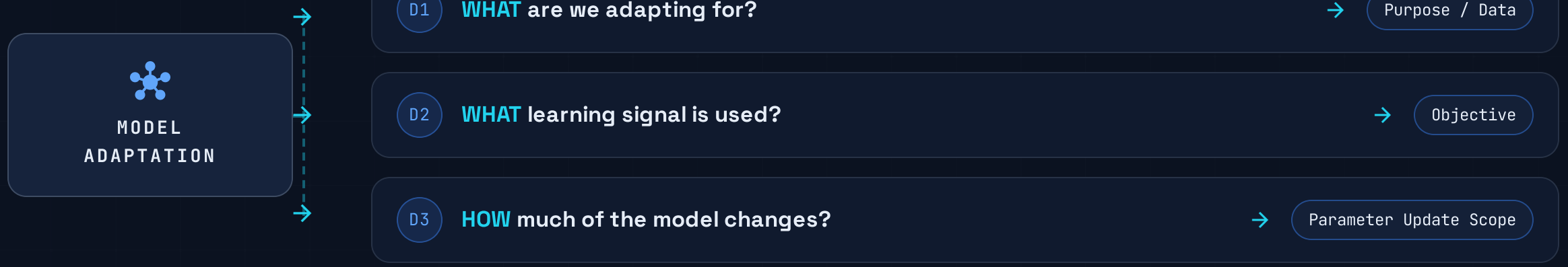
| METRIC | BASE<br>STEP 2 BASELINE | FINE-TUNED<br>AFTER FINE-TUNING | |
|---|---|---|---|
| Accuracy | 68% | **91%** | +23 pts |
| Macro-F1 | 0.64 | **0.89** | +0.25 |
| Valid structured output | 71% | **98%** | +27 pts |
| Avg. latency small but real | 120 ms | 125 ms | +5 ms |

Did the quality improvement justify the additional complexity?

This is an **engineering decision**, not merely an ML metric decision.

# Types of Fine-Tuning — A Three-Dimensional Taxonomy

There is no single flat list. Fine-tuning can be classified along **different dimensions** — and the categories are **not mutually exclusive**.



# DIMENSION 1 · Adaptation Goal / Data

**What capability or distribution are we adapting to?** These describe the goal or nature of the adaptation dataset.

## QUADRANT 1A
### Task-Specific

Improve **one clearly defined downstream task**.

`sentiment classification` `spam detection` `student-query routing` `named-entity recognition` `medical-text classification`

Best when the target task is **clearly measurable**.

## QUADRANT 1B
### Domain Adaptation

`General Pre-trained LLM + Large Domain Corpus → Continue Language-Model Training → Domain-Adapted Model`

`medical corpus → medically adapted` · `legal corpus → legally adapted`

Terminology — also called **Domain-Adaptive Pre-Training (DAPT)** / continued pre-training.

**NOT necessarily SFT** — the objective may still be language modeling.

## QUADRANT 1C
### Instruction Tuning

`Instruction: Summarize this paragraph → Response`

`Instruction: Translate this sentence → Response`

`Instruction: Extract the invoice number → Response`

Purpose — improve ability to **follow natural-language instructions**.

Often implemented using **supervised fine-tuning** — not mutually exclusive with SFT.

## QUADRANT 1D
### Multi-Task

Training data **mixes** several task types:

`Classification` + `Summarization` + `Extraction` + `Question Answering` + `Generation`

**Benefit:** shared representations / generalization.

**Challenge:** task interference and dataset imbalance.

# DIMENSION 2 · Training Signal / Objective

What learning signal tells the model how it should change?

TRAINING OBJECTIVE

- ▸ Supervised targets
- ▸ Language-model objective
- ▸ Preference comparisons
- ▸ Reward / RL-based signal

This is a different classification axis from task / domain.

## SFT

SUPERVISED FINE-TUNING

Fine-tuning using examples where **the desired output is known**.

```
Input 'My hall ticket is unavailable.' → Target 'Department: Examination / Priority: High'
```

```
Prediction → Compare with Target → Loss → Backpropagation
```

One of the most common LLM post-training stages.

## Continued LM Training

SAME SIGNAL · NEW CORPUS

Signal remains **Previous tokens → Predict next token** — but the **corpus changes**.

```
General LLM + large cybersecurity corpus → continued LM training → Cybersecurity-adapted model
```

Different from providing explicit instruction-target pairs.

## Preference / Alignment Optimization

LEARNING FROM COMPARISONS

```
Prompt → Response A + Response B → A is preferred to B
```

RLHF · DPO · other preference-optimization methods

Purposes — preferred response behavior · helpfulness · stylistic preferences · policy / alignment objectives.

Keep algorithms conceptual here.

# DIMENSION 3 · Parameter-Update Scope

Which model parameters are actually allowed to change?

SCOPE 3·A

## Full Fine-Tuning

| | |
|---|---|
| Layer 1 | TRAINABLE |
| Layer 2 | TRAINABLE |
| Layer 3 | TRAINABLE |
| Layer 4 | TRAINABLE |
| Layer N | TRAINABLE |

Most/all model parameters are updated.

**BENEFITS**

Maximum adaptation flexibility.

**COSTS**

High memory · high compute · large task-specific checkpoints · expensive for large LLMs.

SCOPE 3·B

## Selective / Partial

| | |
|---|---|
| Layer 1 | FROZEN |
| Layer 2 | FROZEN |
| Layer 3 | FROZEN |
| Layer 4 | TRAINABLE |
| Layer 5 | TRAINABLE |

Main idea — update a **subset of existing model parameters**.

**STRATEGIES**

Strategies — train final layers only · train task head · train selected parameter groups.

SCOPE 3·C

## PEFT

| | |
|---|---|
| Layer 1 | FROZEN |
| Layer 2 | FROZEN |
| Layers 3 - N-1 | FROZEN |
| Layer N | FROZEN |
| ADAPTER | TRAINABLE |

Most base parameters **frozen**, small adaptation trained.

**Preview only** — no LoRA mathematics yet.

NEXT LECTURE PEFT → LoRA → QLoRA

SECTION 03 · TAXONOMY

TAXONOMY

# The Correct Fine-Tuning Taxonomy

| DIMENSION | EXAMPLE CATEGORIES | MAIN QUESTION |
|---|---|---|
| D1 **Adaptation goal / data** | Task-specific · domain adaptation · instruction tuning · multi-task | *What are we adapting for?* |
| D2 **Training signal / objective** | SFT · continued LM objective · preference optimization / RLHF | *How is learning supervised?* |
| D3 **Parameter-update scope** | Full FT · selective FT · PEFT | *How much of the model changes?* |

These categories **overlap** .

## ONE MODEL CAN BELONG TO SEVERAL CATEGORIES

EXAMPLE · UNIVERSITY HELPDESK EXPERIMENT

PURPOSE
Task-specific

+

TRAINING SIGNAL
Supervised Fine-Tuning

+

FORMAT
Instruction / chat examples

+

PARAMETER UPDATE
Full FT or LoRA

Task-Specific + Supervised + Instruction-Formatted + LoRA-Based — **there is no contradiction.**

# Terminology Traps & Classification Drill

## Common Terminology Mistakes

MYTH → REALITY

✕ *"SFT and instruction tuning are completely different competing choices."*

✓ **Not necessarily** — instruction tuning is commonly performed using SFT.

---

✕ *"Domain adaptation always means instruction-response training."*

✓ **No** — it may use continued language-model pre-training.

---

✕ *"LoRA is another task type."*

✓ **No** — LoRA describes how parameters are adapted, not what task is being learned.

---

✕ *"Task-specific FT automatically means full fine-tuning."*

✓ **No** — task-specific adaptation can use full FT, selective FT or PEFT.

## Classroom Activity — Name the Dimensions

ANSWER BEFORE THE REVEAL

1. Sentiment classification using labeled input-output examples, updating all weights
   → Goal: Task-specific · Signal: SFT · Strategy: Full FT

2. Continue training an LLM on millions of legal documents
   → Goal: Domain adaptation · Signal: LM objective · Strategy: depends on implementation

3. Train thousands of instruction-response examples using LoRA
   → Goal: Instruction tuning · Signal: SFT · Strategy: PEFT / LoRA

4. Optimize using preferred and rejected responses
   → Signal: Preference optimization · Could be DPO / RLHF-style

SECTION 04 · WRAP-UP

SUMMARY

# Summary & Bridge to Lecture 4

## The Fine-Tuning Process

ONE CONNECTED PIPELINE

14 STEPS

01 Define Task
02 Baseline
03 Base Model
04 Dataset
05 Clean / Label
06 Train / Val / Test
07 Format / Tokenize
08 Forward Pass
09 Loss
10 Backpropagation
11 Optimizer Update
12 Validation
13 Test
14 Deployment

## The Taxonomy

THREE QUESTIONS

3 AXES

WHAT? → Task / Domain / Instruction / Multi-Task

HOW LEARN? → SFT / LM Objective / Preferences

WHAT CHANGES? → Full / Selective / PEFT

BRIDGE TO LECTURE 4

`7B Parameter Model + Full Fine-Tuning → Most/all 7B parameters trainable → High training memory · High compute · Large checkpoint`

**Do we really need to update all seven billion parameters to specialize the model?**

LECTURE 4: PEFT → LoRA → QLoRA